

Q1 - Company Employee & Department Database

Q1.A1: INNER JOIN - emp name, dept name, designation

```
●●● Q1.A1: INNER JOIN - emp name, dept name, designation

mysql> USE q1_company;
Database changed

mysql> SELECT e.emp_name, d.dept_name, e.designation
-> FROM EMPLOYEE e
-> INNER JOIN DEPARTMENT d ON e.dept_id = d.dept_id;

+-----+-----+-----+
| emp_name | dept_name | designation |
+-----+-----+-----+
| Arjun Mehta | Engineering | Senior Engineer |
| Divya Rao | Engineering | Engineer |
| Kiran Shah | Engineering | Engineer |
| Neha Kulkarni | Finance | Finance Manager |
| Rohit Verma | Finance | Accountant |
| Sneha Iyer | Marketing | Marketing Lead |
| Aditya Nair | Marketing | Executive |
| Priya Menon | Sales | Sales Manager |
| Vikram Singh | Sales | Sales Executive |
| Anjali Gupta | Support | Support Engineer |
+-----+-----+-----+

10 rows in set (0.00 sec)
```

Q1.A2: LEFT JOIN - all departments with employees (incl. empty depts)

```
●●● Q1.A2: LEFT JOIN - all departments with employees (incl. empty depts)

mysql> USE q1_company;
Database changed

mysql> SELECT d.dept_name, e.emp_name
-> FROM DEPARTMENT d
-> LEFT JOIN EMPLOYEE e ON d.dept_id = e.dept_id
-> ORDER BY d.dept_name;

+-----+-----+
| dept_name | emp_name |
+-----+-----+
| Engineering | Arjun Mehta |
| Engineering | Divya Rao |
| Engineering | Kiran Shah |
| Finance | Neha Kulkarni |
| Finance | Rohit Verma |
| HR | NULL |
| Marketing | Sneha Iyer |
| Marketing | Aditya Nair |
| Sales | Priya Menon |
| Sales | Vikram Singh |
| Support | Anjali Gupta |
+-----+-----+

11 rows in set (0.00 sec)
```

Q1.A3: RIGHT JOIN - all employees with dept (incl. unassigned)

Q1.A3: RIGHT JOIN - all employees with dept (incl. unassigned)

```
mysql> USE q1_company;
Database changed

mysql> SELECT e.emp_name, d.dept_name
-> FROM DEPARTMENT d
-> RIGHT JOIN EMPLOYEE e ON d.dept_id = e.dept_id
-> ORDER BY e.emp_name;
+-----+-----+
| emp_name | dept_name |
+-----+-----+
| Aditya Nair | Marketing |
| Anjali Gupta | Support   |
| Arjun Mehta | Engineering |
| Divya Rao   | Engineering |
| Karan Malhotra | NULL      |
| Kiran Shah  | Engineering |
| Meera Pillai | NULL      |
| Neha Kulkarni | Finance   |
| Priya Menon | Sales     |
| Rohit Verma | Finance   |
| Sneha Iyer  | Marketing |
| Vikram Singh | Sales     |
+-----+-----+
12 rows in set (0.00 sec)
```

Q1.A4: SELF JOIN - employee & manager name

Q1.A4: SELF JOIN - employee & manager name

```
mysql> USE q1_company;
Database changed

mysql> SELECT e.emp_name AS employee, m.emp_name AS manager
-> FROM EMPLOYEE e
-> LEFT JOIN EMPLOYEE m ON e.manager_id = m.emp_id
-> ORDER BY e.emp_name;
+-----+-----+
| employee | manager |
+-----+-----+
| Aditya Nair | Sneha Iyer |
| Anjali Gupta | NULL      |
| Arjun Mehta | NULL      |
| Divya Rao   | Arjun Mehta |
| Karan Malhotra | NULL      |
| Kiran Shah  | Arjun Mehta |
| Meera Pillai | NULL      |
| Neha Kulkarni | NULL      |
| Priya Menon | NULL      |
| Rohit Verma | Neha Kulkarni |
| Sneha Iyer  | NULL      |
| Vikram Singh | Priya Menon |
+-----+-----+
12 rows in set (0.00 sec)
```

Q1.B1: VIEW - EMP_DEPT_VIEW

```
Q1.B1: VIEW - EMP_DEPT_VIEW

mysql> USE q1_company;
Database changed

mysql> DROP VIEW IF EXISTS EMP_DEPT_VIEW;
Query OK, 0 rows affected (0.00 sec)

mysql> CREATE VIEW EMP_DEPT_VIEW AS
-> SELECT e.emp_name, d.dept_name, e.designation, e.salary
-> FROM EMPLOYEE e
-> LEFT JOIN DEPARTMENT d ON e.dept_id = d.dept_id;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM EMP_DEPT_VIEW;
+-----+-----+-----+-----+
| emp_name | dept_name | designation | salary |
+-----+-----+-----+-----+
| Arjun Mehta | Engineering | Senior Engineer | 75000 |
| Divya Rao | Engineering | Engineer | 58000 |
| Kiran Shah | Engineering | Engineer | 52000 |
| Neha Kulkarni | Finance | Finance Manager | 82000 |
| Rohit Verma | Finance | Accountant | 45000 |
| Sneha Iyer | Marketing | Marketing Lead | 68000 |
| Aditya Nair | Marketing | Executive | 41000 |
| Priya Menon | Sales | Sales Manager | 71000 |
| Vikram Singh | Sales | Sales Executive | 39000 |
| Anjali Gupta | Support | Support Engineer | 47000 |
| Karan Malhotra | NULL | Consultant | 60000 |
| Meera Pillai | NULL | Freelancer | 55000 |
+-----+-----+-----+-----+
12 rows in set (0.00 sec)
```

Q1.B2: VIEW built on VIEW - HIGH_EARNERS

```
Q1.B2: VIEW built on VIEW - HIGH_EARNERS

mysql> USE q1_company;
Database changed

mysql> DROP VIEW IF EXISTS HIGH_EARNERS;
Query OK, 0 rows affected (0.00 sec)

mysql> CREATE VIEW HIGH_EARNERS AS
-> SELECT emp_name, dept_name, designation, salary
-> FROM EMP_DEPT_VIEW
-> WHERE salary > 50000;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM HIGH_EARNERS ORDER BY salary DESC;
+-----+-----+-----+-----+
| emp_name | dept_name | designation | salary |
+-----+-----+-----+-----+
| Neha Kulkarni | Finance | Finance Manager | 82000 |
| Arjun Mehta | Engineering | Senior Engineer | 75000 |
| Priya Menon | Sales | Sales Manager | 71000 |
| Sneha Iyer | Marketing | Marketing Lead | 68000 |
| Karan Malhotra | NULL | Consultant | 60000 |
| Divya Rao | Engineering | Engineer | 58000 |
| Meera Pillai | NULL | Freelancer | 55000 |
| Kiran Shah | Engineering | Engineer | 52000 |
+-----+-----+-----+-----+
8 rows in set (0.00 sec)
```

Q1.B3: VIEW - DEPT_PROJECT_COUNT

```
Q1.B3: VIEW - DEPT_PROJECT_COUNT

mysql> USE q1_company;
Database changed

mysql> DROP VIEW IF EXISTS DEPT_PROJECT_COUNT;
Query OK, 0 rows affected (0.00 sec)

mysql> CREATE VIEW DEPT_PROJECT_COUNT AS
-> SELECT d.dept_name, COUNT(p.proj_id) AS project_count
-> FROM DEPARTMENT d
-> LEFT JOIN PROJECT p ON d.dept_id = p.dept_id
-> GROUP BY d.dept_name;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM DEPT_PROJECT_COUNT ORDER BY project_count DESC;
+-----+-----+
| dept_name | project_count |
+-----+-----+
| Engineering | 2 |
| Finance | 1 |
| Marketing | 1 |
| Sales | 1 |
| HR | 0 |
| Support | 0 |
+-----+-----+
6 rows in set (0.00 sec)
```

Q2 - University Student, Course & Faculty Database

Q2.A1: INNER JOIN - student, course, marks

```
Q2.A1: INNER JOIN - student, course, marks

mysql> USE q2_university;
Database changed

mysql> SELECT s.student_name, c.course_name, en.marks
-> FROM STUDENT s
-> INNER JOIN ENROLLMENT en ON s.student_id = en.student_id
-> INNER JOIN COURSE c ON en.course_id = c.course_id
-> ORDER BY s.student_name;

+-----+-----+-----+
| student_name | course_name | marks |
+-----+-----+-----+
| Aarav Sharma | Database Systems | 85 |
| Aarav Sharma | Data Structures | 92 |
| Bhavya Reddy | Database Systems | 78 |
| Bhavya Reddy | Digital Electronics | 65 |
| Bhavya Reddy | Data Structures | 68 |
| Chetan Joshi | Digital Electronics | 88 |
| Chetan Joshi | Thermodynamics | 55 |
| Diya Kapoor | Digital Electronics | 95 |
| Diya Kapoor | Signals & Systems | 70 |
| Esha Bansal | Thermodynamics | 60 |
| Esha Bansal | Signals & Systems | 82 |
| Farhan Ali | Thermodynamics | 48 |
| Farhan Ali | Signals & Systems | 74 |
| Gauri Desai | Database Systems | 91 |
| Gauri Desai | Data Structures | 76 |
+-----+-----+-----+
15 rows in set (0.00 sec)
```

Q2.A2: LEFT JOIN - all students with courses (incl. none enrolled)

```
Q2.A2: LEFT JOIN - all students with courses (incl. none enrolled)

mysql> USE q2_university;
Database changed

mysql> SELECT s.student_name, c.course_name
-> FROM STUDENT s
-> LEFT JOIN ENROLLMENT en ON s.student_id = en.student_id
-> LEFT JOIN COURSE c ON en.course_id = c.course_id
-> ORDER BY s.student_name;

+-----+-----+
| student_name | course_name |
+-----+-----+
| Aarav Sharma | Database Systems |
| Aarav Sharma | Data Structures |
| Bhavya Reddy | Database Systems |
| Bhavya Reddy | Digital Electronics |
| Bhavya Reddy | Data Structures |
| Chetan Joshi | Digital Electronics |
| Chetan Joshi | Thermodynamics |
| Diya Kapoor | Digital Electronics |
| Diya Kapoor | Signals & Systems |
| Esha Bansal | Thermodynamics |
| Esha Bansal | Signals & Systems |
| Farhan Ali | Thermodynamics |
| Farhan Ali | Signals & Systems |
| Gauri Desai | Database Systems |
| Gauri Desai | Data Structures |
| Ishaan Trivedi | NULL |
+-----+-----+
16 rows in set (0.00 sec)
```


Q2.A3: RIGHT JOIN - all courses with students (incl. zero enrollment)

●●● Q2.A3: RIGHT JOIN - all courses with students (incl. zero enrollment)

```
mysql> USE q2_university;
Database changed

mysql> SELECT c.course_name, s.student_name
-> FROM ENROLLMENT en
-> RIGHT JOIN COURSE c ON en.course_id = c.course_id
-> LEFT JOIN STUDENT s ON en.student_id = s.student_id
-> ORDER BY c.course_name;

+-----+-----+
| course_name | student_name |
+-----+-----+
| Data Structures | Aarav Sharma |
| Data Structures | Gauri Desai |
| Data Structures | Bhavya Reddy |
| Database Systems | Aarav Sharma |
| Database Systems | Bhavya Reddy |
| Database Systems | Gauri Desai |
| Digital Electronics | Bhavya Reddy |
| Digital Electronics | Chetan Joshi |
| Digital Electronics | Diya Kapoor |
| Robotics | NULL |
| Signals & Systems | Diya Kapoor |
| Signals & Systems | Esha Bansal |
| Signals & Systems | Farhan Ali |
| Thermodynamics | Chetan Joshi |
| Thermodynamics | Esha Bansal |
| Thermodynamics | Farhan Ali |
+-----+-----+
16 rows in set (0.00 sec)
```

Q2.A4: Faculty whose courses have NO student scoring above 80

●●● Q2.A4: Faculty whose courses have NO student scoring above 80

```
mysql> USE q2_university;
Database changed

mysql> SELECT f.faculty_name, c.course_name, MAX(en.marks) AS highest_marks
-> FROM FACULTY f
-> JOIN COURSE c ON f.faculty_id = c.faculty_id
-> LEFT JOIN ENROLLMENT en ON c.course_id = en.course_id
-> GROUP BY f.faculty_name, c.course_name
-> HAVING MAX(en.marks) <= 80 OR MAX(en.marks) IS NULL;

+-----+-----+-----+
| faculty_name | course_name | highest_marks |
+-----+-----+-----+
| Dr. Manoj Pillai | Thermodynamics | 60 |
| Dr. Manoj Pillai | Robotics | NULL |
+-----+-----+-----+
2 rows in set (0.00 sec)
```

Q2.B1: VIEW - STUDENT_COURSE_VIEW

```
Q2.B1: VIEW - STUDENT_COURSE_VIEW

mysql> USE q2_university;
Database changed

mysql> DROP VIEW IF EXISTS STUDENT_COURSE_VIEW;
Query OK, 0 rows affected (0.00 sec)

mysql> CREATE VIEW STUDENT_COURSE_VIEW AS
-> SELECT s.student_name, c.course_name, f.faculty_name, en.marks
-> FROM ENROLLMENT en
-> JOIN STUDENT s ON en.student_id = s.student_id
-> JOIN COURSE c ON en.course_id = c.course_id
-> JOIN FACULTY f ON c.faculty_id = f.faculty_id;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM STUDENT_COURSE_VIEW ORDER BY student_name;
+-----+-----+-----+-----+
| student_name | course_name      | faculty_name      | marks |
+-----+-----+-----+-----+
| Aarav Sharma | Database Systems | Dr. Ramesh Iyer   | 85    |
| Aarav Sharma | Data Structures  | Dr. Kavita Nambiar | 92    |
| Bhavya Reddy | Database Systems | Dr. Ramesh Iyer   | 78    |
| Bhavya Reddy | Digital Electronics | Dr. Sunita Rao    | 65    |
| Bhavya Reddy | Data Structures  | Dr. Kavita Nambiar | 68    |
| Chetan Joshi | Digital Electronics | Dr. Sunita Rao    | 88    |
| Chetan Joshi | Thermodynamics   | Dr. Manoj Pillai  | 55    |
| Diya Kapoor  | Digital Electronics | Dr. Sunita Rao    | 95    |
| Diya Kapoor  | Signals & Systems | Dr. Arvind Menon  | 70    |
| Esha Bansal  | Thermodynamics   | Dr. Manoj Pillai  | 60    |
| Esha Bansal  | Signals & Systems | Dr. Arvind Menon  | 82    |
| Farhan Ali   | Thermodynamics   | Dr. Manoj Pillai  | 48    |
| Farhan Ali   | Signals & Systems | Dr. Arvind Menon  | 74    |
| Gauri Desai  | Database Systems | Dr. Ramesh Iyer   | 91    |
| Gauri Desai  | Data Structures  | Dr. Kavita Nambiar | 76    |
+-----+-----+-----+-----+
15 rows in set (0.00 sec)
```

Q2.B2: VIEW built on VIEW - TOPPERS (marks > 90)

Q2.B2: VIEW built on VIEW - TOPPERS (marks > 90)

```
mysql> USE q2_university;
Database changed

mysql> DROP VIEW IF EXISTS TOPPERS;
Query OK, 0 rows affected (0.00 sec)

mysql> CREATE VIEW TOPPERS AS
-> SELECT student_name, course_name, faculty_name, marks
-> FROM STUDENT_COURSE_VIEW
-> WHERE marks > 90;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM TOPPERS ORDER BY marks DESC;
+-----+-----+-----+-----+
| student_name | course_name      | faculty_name      | marks |
+-----+-----+-----+-----+
| Diya Kapoor  | Digital Electronics | Dr. Sunita Rao    | 95    |
| Aarav Sharma | Data Structures  | Dr. Kavita Nambiar | 92    |
| Gauri Desai  | Database Systems | Dr. Ramesh Iyer   | 91    |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

Q2.B3: VIEW - FACULTY_WORKLOAD (GROUP BY inside view)

```
●●● Q2.B3: VIEW - FACULTY_WORKLOAD (GROUP BY inside view)

mysql> USE q2_university;
Database changed

mysql> DROP VIEW IF EXISTS FACULTY_WORKLOAD;
Query OK, 0 rows affected (0.00 sec)

mysql> CREATE VIEW FACULTY_WORKLOAD AS
-> SELECT f.faculty_name, COUNT(c.course_id) AS num_courses
-> FROM FACULTY f
-> LEFT JOIN COURSE c ON f.faculty_id = c.faculty_id
-> GROUP BY f.faculty_name;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM FACULTY_WORKLOAD ORDER BY num_courses DESC;
+-----+-----+
| faculty_name | num_courses |
+-----+-----+
| Dr. Manoj Pillai | 2 |
| Dr. Ramesh Iyer | 1 |
| Dr. Sunita Rao | 1 |
| Dr. Kavita Nambiar | 1 |
| Dr. Arvind Menon | 1 |
+-----+-----+
5 rows in set (0.00 sec)
```

Q2.B4: Faculty with MAX workload (query on the view)

```
●●● Q2.B4: Faculty with MAX workload (query on the view)

mysql> USE q2_university;
Database changed

mysql> SELECT faculty_name, num_courses
-> FROM FACULTY_WORKLOAD
-> WHERE num_courses = (SELECT MAX(num_courses) FROM FACULTY_WORKLOAD);
+-----+-----+
| faculty_name | num_courses |
+-----+-----+
| Dr. Manoj Pillai | 2 |
+-----+-----+
1 row in set (0.00 sec)
```


Q3 - Hospital Patient, Doctor & Appointment Database

Q3.A1: INNER JOIN - patient, doctor, appt_date

```
●●● Q3.A1: INNER JOIN - patient, doctor, appt_date

mysql> USE q3_hospital;
Database changed

mysql> SELECT p.patient_name, doc.doctor_name, a.appt_date
-> FROM PATIENT p
-> INNER JOIN APPOINTMENT a ON p.patient_id = a.patient_id
-> INNER JOIN DOCTOR doc ON a.doctor_id = doc.doctor_id
-> ORDER BY a.appt_date;
+-----+-----+-----+
| patient_name | doctor_name | appt_date |
+-----+-----+-----+
| Amit Sharma  | Dr. Anil Kapoor | 2026-01-05 |
| Sunita Verma | Dr. Anil Kapoor | 2026-01-06 |
| Rakesh Yadav | Dr. Rajeev Menon | 2026-01-07 |
| Pooja Agarwal | Dr. Fatima Sheikh | 2026-01-08 |
| Manoj Tiwari | Dr. Fatima Sheikh | 2026-01-09 |
| Kavya Reddy  | Dr. Ayesha Khan | 2026-01-10 |
| Sameer Khan  | Dr. Vivek Nair  | 2026-01-11 |
| Nisha Joshi  | Dr. Rajeev Menon | 2026-01-12 |
| Ramesh Gowda | Dr. Lata Choudhary | 2026-01-13 |
| Amit Sharma  | Dr. Suresh Pillai | 2026-01-14 |
| Sunita Verma | Dr. Ayesha Khan | 2026-01-15 |
| Rakesh Yadav | Dr. Lata Choudhary | 2026-01-16 |
| Pooja Agarwal | Dr. Anil Kapoor | 2026-01-17 |
| Manoj Tiwari | Dr. Vivek Nair  | 2026-01-18 |
| Kavya Reddy  | Dr. Suresh Pillai | 2026-01-19 |
+-----+-----+-----+
15 rows in set (0.00 sec)
```

Q3.A2: LEFT JOIN - all patients with appointments (incl. none)

```
●●● Q3.A2: LEFT JOIN - all patients with appointments (incl. none)

mysql> USE q3_hospital;
Database changed

mysql> SELECT p.patient_name, a.appt_date, a.status
-> FROM PATIENT p
-> LEFT JOIN APPOINTMENT a ON p.patient_id = a.patient_id
-> ORDER BY p.patient_name;
+-----+-----+-----+
| patient_name | appt_date | status |
+-----+-----+-----+
| Amit Sharma  | 2026-01-14 | Pending |
| Amit Sharma  | 2026-01-05 | Completed |
| Kavya Reddy  | 2026-01-19 | Pending |
| Kavya Reddy  | 2026-01-10 | Cancelled |
| Manoj Tiwari | 2026-01-18 | Cancelled |
| Manoj Tiwari | 2026-01-09 | Completed |
| Nisha Joshi  | 2026-01-12 | Pending |
| Pooja Agarwal | 2026-01-17 | Completed |
| Pooja Agarwal | 2026-01-08 | Pending |
| Rakesh Yadav | 2026-01-16 | Pending |
| Rakesh Yadav | 2026-01-07 | Completed |
| Ramesh Gowda | 2026-01-13 | Completed |
| Rohan Malhotra | NULL | NULL |
| Sameer Khan  | 2026-01-11 | Completed |
| Sunita Verma | 2026-01-15 | Completed |
| Sunita Verma | 2026-01-06 | Pending |
+-----+-----+-----+
16 rows in set (0.00 sec)
```

Q3.A3: RIGHT JOIN - all doctors with appointments (incl. none)

```
Q3.A3: RIGHT JOIN - all doctors with appointments (incl. none)

mysql> USE q3_hospital;
Database changed

mysql> SELECT doc.doctor_name, a.appt_date, a.status
-> FROM APPOINTMENT a
-> RIGHT JOIN DOCTOR doc ON a.doctor_id = doc.doctor_id
-> ORDER BY doc.doctor_name;

+-----+-----+-----+
| doctor_name | appt_date | status |
+-----+-----+-----+
| Dr. Anil Kapoor | 2026-01-17 | Completed |
| Dr. Anil Kapoor | 2026-01-06 | Pending |
| Dr. Anil Kapoor | 2026-01-05 | Completed |
| Dr. Ayesha Khan | 2026-01-15 | Completed |
| Dr. Ayesha Khan | 2026-01-10 | Cancelled |
| Dr. Fatima Sheikh | 2026-01-09 | Completed |
| Dr. Fatima Sheikh | 2026-01-08 | Pending |
| Dr. Lata Choudhary | 2026-01-16 | Pending |
| Dr. Lata Choudhary | 2026-01-13 | Completed |
| Dr. Rajeev Menon | 2026-01-12 | Pending |
| Dr. Rajeev Menon | 2026-01-07 | Completed |
| Dr. Sarita Bose | NULL | NULL |
| Dr. Suresh Pillai | 2026-01-19 | Pending |
| Dr. Suresh Pillai | 2026-01-14 | Pending |
| Dr. Vivek Nair | 2026-01-18 | Cancelled |
| Dr. Vivek Nair | 2026-01-11 | Completed |
+-----+-----+-----+
16 rows in set (0.00 sec)
```

Q3.A4: Doctor, specialization, department

```
Q3.A4: Doctor, specialization, department

mysql> USE q3_hospital;
Database changed

mysql> SELECT doc.doctor_name, doc.specialization, d.dept_name
-> FROM DOCTOR doc
-> JOIN DEPARTMENT d ON doc.dept_id = d.dept_id
-> ORDER BY d.dept_name;

+-----+-----+-----+
| doctor_name | specialization | dept_name |
+-----+-----+-----+
| Dr. Anil Kapoor | Cardiologist | Cardiology |
| Dr. Sarita Bose | Cardiologist | Cardiology |
| Dr. Vivek Nair | Dermatologist | Dermatology |
| Dr. Fatima Sheikh | Neurologist | Neurology |
| Dr. Suresh Pillai | Neurologist | Neurology |
| Dr. Rajeev Menon | Orthopedic | Orthopedics |
| Dr. Lata Choudhary | Orthopedic | Orthopedics |
| Dr. Ayesha Khan | Pediatrician | Pediatrics |
+-----+-----+-----+
8 rows in set (0.00 sec)
```

Q3.B1: VIEW - APPOINTMENT_DETAILS

```
Q3.B1: VIEW - APPOINTMENT_DETAILS

mysql> USE q3_hospital;
Database changed

mysql> DROP VIEW IF EXISTS APPOINTMENT_DETAILS;
Query OK, 0 rows affected (0.00 sec)

mysql> CREATE VIEW APPOINTMENT_DETAILS AS
-> SELECT a.appt_id, p.patient_name, doc.doctor_name, doc.specialization,
->        a.appt_date, a.status
-> FROM APPOINTMENT a
-> JOIN PATIENT p ON a.patient_id = p.patient_id
-> JOIN DOCTOR doc ON a.doctor_id = doc.doctor_id;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM APPOINTMENT_DETAILS ORDER BY appt_date;
+-----+-----+-----+-----+-----+-----+
| appt_id | patient_name | doctor_name | specialization | appt_date | status |
+-----+-----+-----+-----+-----+-----+
| 501 | Amit Sharma | Dr. Anil Kapoor | Cardiologist | 2026-01-05 | Completed |
| 502 | Sunita Verma | Dr. Anil Kapoor | Cardiologist | 2026-01-06 | Pending |
| 503 | Rakesh Yadav | Dr. Rajeev Menon | Orthopedic | 2026-01-07 | Completed |
| 504 | Pooja Agarwal | Dr. Fatima Sheikh | Neurologist | 2026-01-08 | Pending |
| 505 | Manoj Tiwari | Dr. Fatima Sheikh | Neurologist | 2026-01-09 | Completed |
| 506 | Kavya Reddy | Dr. Ayesha Khan | Pediatrician | 2026-01-10 | Cancelled |
| 507 | Sameer Khan | Dr. Vivek Nair | Dermatologist | 2026-01-11 | Completed |
| 508 | Nisha Joshi | Dr. Rajeev Menon | Orthopedic | 2026-01-12 | Pending |
| 509 | Ramesh Gowda | Dr. Lata Choudhary | Orthopedic | 2026-01-13 | Completed |
| 510 | Amit Sharma | Dr. Suresh Pillai | Neurologist | 2026-01-14 | Pending |
| 511 | Sunita Verma | Dr. Ayesha Khan | Pediatrician | 2026-01-15 | Completed |
| 512 | Rakesh Yadav | Dr. Lata Choudhary | Orthopedic | 2026-01-16 | Pending |
| 513 | Pooja Agarwal | Dr. Anil Kapoor | Cardiologist | 2026-01-17 | Completed |
| 514 | Manoj Tiwari | Dr. Vivek Nair | Dermatologist | 2026-01-18 | Cancelled |
| 515 | Kavya Reddy | Dr. Suresh Pillai | Neurologist | 2026-01-19 | Pending |
+-----+-----+-----+-----+-----+-----+
15 rows in set (0.00 sec)
```

Q3.B2: VIEW - DOCTOR_APPOINTMENT_COUNT

```
Q3.B2: VIEW - DOCTOR_APPOINTMENT_COUNT

mysql> USE q3_hospital;
Database changed

mysql> DROP VIEW IF EXISTS DOCTOR_APPOINTMENT_COUNT;
Query OK, 0 rows affected (0.00 sec)

mysql> CREATE VIEW DOCTOR_APPOINTMENT_COUNT AS
-> SELECT doc.doctor_name, COUNT(a.appt_id) AS total_appointments
-> FROM DOCTOR doc
-> LEFT JOIN APPOINTMENT a ON doc.doctor_id = a.doctor_id
-> GROUP BY doc.doctor_name;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM DOCTOR_APPOINTMENT_COUNT ORDER BY total_appointments DESC;
+-----+-----+
| doctor_name | total_appointments |
+-----+-----+
| Dr. Anil Kapoor | 3 |
| Dr. Rajeev Menon | 2 |
| Dr. Fatima Sheikh | 2 |
| Dr. Suresh Pillai | 2 |
| Dr. Ayesha Khan | 2 |
| Dr. Vivek Nair | 2 |
| Dr. Lata Choudhary | 2 |
| Dr. Sarita Bose | 0 |
+-----+-----+
8 rows in set (0.00 sec)
```

Q3.B3: VIEW - PENDING_APPOINTMENTS (based on

APPOINTMENT_DETAILS)

```
●●● Q3.B3: VIEW - PENDING_APPOINTMENTS (based on APPOINTMENT_DETAILS)

mysql> USE q3_hospital;
Database changed

mysql> DROP VIEW IF EXISTS PENDING_APPOINTMENTS;
Query OK, 0 rows affected (0.00 sec)

mysql> CREATE VIEW PENDING_APPOINTMENTS AS
  -> SELECT appt_id, patient_name, doctor_name, specialization, appt_date, status
  -> FROM APPOINTMENT_DETAILS
  -> WHERE status = 'Pending';
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM PENDING_APPOINTMENTS ORDER BY appt_date;
+-----+-----+-----+-----+-----+-----+
| appt_id | patient_name | doctor_name | specialization | appt_date | status |
+-----+-----+-----+-----+-----+-----+
| 502 | Sunita Verma | Dr. Anil Kapoor | Cardiologist | 2026-01-06 | Pending |
| 504 | Pooja Agarwal | Dr. Fatima Sheikh | Neurologist | 2026-01-08 | Pending |
| 508 | Nisha Joshi | Dr. Rajeev Menon | Orthopedic | 2026-01-12 | Pending |
| 510 | Amit Sharma | Dr. Suresh Pillai | Neurologist | 2026-01-14 | Pending |
| 512 | Rakesh Yadav | Dr. Lata Choudhary | Orthopedic | 2026-01-16 | Pending |
| 515 | Kavya Reddy | Dr. Suresh Pillai | Neurologist | 2026-01-19 | Pending |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

Q3.B4: Update via BASE TABLE, verify through the view

```
●●● Q3.B4: Update via BASE TABLE, verify through the view

mysql> USE q3_hospital;
Database changed

-- BEFORE: show pending list (appt_id 502 present)
mysql> SELECT * FROM PENDING_APPOINTMENTS WHERE appt_id = 502;
+-----+-----+-----+-----+-----+-----+
| appt_id | patient_name | doctor_name | specialization | appt_date | status |
+-----+-----+-----+-----+-----+-----+
| 502 | Sunita Verma | Dr. Anil Kapoor | Cardiologist | 2026-01-06 | Pending |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

-- Update the base table
mysql> UPDATE APPOINTMENT SET status = 'Completed' WHERE appt_id = 502;
Query OK, 1 row affected (0.00 sec)

-- AFTER: verify the view reflects the change (502 should disappear)
mysql> SELECT * FROM PENDING_APPOINTMENTS ORDER BY appt_date;
+-----+-----+-----+-----+-----+-----+
| appt_id | patient_name | doctor_name | specialization | appt_date | status |
+-----+-----+-----+-----+-----+-----+
| 504 | Pooja Agarwal | Dr. Fatima Sheikh | Neurologist | 2026-01-08 | Pending |
| 508 | Nisha Joshi | Dr. Rajeev Menon | Orthopedic | 2026-01-12 | Pending |
| 510 | Amit Sharma | Dr. Suresh Pillai | Neurologist | 2026-01-14 | Pending |
| 512 | Rakesh Yadav | Dr. Lata Choudhary | Orthopedic | 2026-01-16 | Pending |
| 515 | Kavya Reddy | Dr. Suresh Pillai | Neurologist | 2026-01-19 | Pending |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```


Q4 - E-Commerce Customer, Order & Product Database

Q4.A1: INNER JOIN across all 4 tables

```
●●● Q4.A1: INNER JOIN across all 4 tables

mysql> USE q4_ecommerce;
Database changed

mysql> SELECT c.cust_name, o.order_date, p.prod_name, oi.quantity
-> FROM CUSTOMER c
-> INNER JOIN ORDERS o ON c.cust_id = o.cust_id
-> INNER JOIN ORDER_ITEMS oi ON o.order_id = oi.order_id
-> INNER JOIN PRODUCT p ON oi.prod_id = p.prod_id
-> ORDER BY c.cust_name, o.order_date;
+-----+-----+-----+-----+
| cust_name | order_date | prod_name | quantity |
+-----+-----+-----+-----+
| Meghna Iyer | 2026-01-12 | Backpack | 1 |
| Meghna Iyer | 2026-01-12 | Bedsheet | 4 |
| Meghna Iyer | 2026-01-30 | Study Table | 2 |
| Meghna Iyer | 2026-01-30 | Smartphone | 1 |
| Priyanka Rao | 2026-01-10 | Laptop | 1 |
| Priyanka Rao | 2026-01-10 | Smartphone | 1 |
| Priyanka Rao | 2026-01-25 | Office Chair | 2 |
| Priyanka Rao | 2026-01-25 | Cookware Set | 1 |
| Rahul Bansal | 2026-01-02 | Laptop | 1 |
| Rahul Bansal | 2026-01-02 | Headphones | 2 |
| Rahul Bansal | 2026-01-15 | Study Table | 1 |
| Rahul Bansal | 2026-01-15 | Office Chair | 1 |
| Sneha Kapoor | 2026-01-05 | Headphones | 1 |
| Sneha Kapoor | 2026-01-05 | Smartphone | 1 |
| Sneha Kapoor | 2026-01-28 | Headphones | 3 |
| Sneha Kapoor | 2026-01-28 | Running Shoes | 1 |
| Vikas Chauhan | 2026-01-08 | Cookware Set | 2 |
| Vikas Chauhan | 2026-01-08 | Bedsheet | 3 |
| Vikas Chauhan | 2026-01-20 | Running Shoes | 1 |
| Vikas Chauhan | 2026-01-20 | Backpack | 2 |
+-----+-----+-----+-----+
20 rows in set (0.00 sec)
```

Q4.A2: LEFT JOIN - all customers with orders (incl. none)

```
●●● Q4.A2: LEFT JOIN - all customers with orders (incl. none)

mysql> USE q4_ecommerce;
Database changed

mysql> SELECT c.cust_name, o.order_id, o.order_date
-> FROM CUSTOMER c
-> LEFT JOIN ORDERS o ON c.cust_id = o.cust_id
-> ORDER BY c.cust_name;
+-----+-----+-----+
| cust_name | order_id | order_date |
+-----+-----+-----+
| Farhan Sheikh | NULL | NULL |
| Meghna Iyer | 8 | 2026-01-12 |
| Meghna Iyer | 10 | 2026-01-30 |
| Priyanka Rao | 6 | 2026-01-10 |
| Priyanka Rao | 7 | 2026-01-25 |
| Rahul Bansal | 1 | 2026-01-02 |
| Rahul Bansal | 2 | 2026-01-15 |
| Sneha Kapoor | 3 | 2026-01-05 |
| Sneha Kapoor | 9 | 2026-01-28 |
| Vikas Chauhan | 4 | 2026-01-08 |
| Vikas Chauhan | 5 | 2026-01-20 |
+-----+-----+-----+
11 rows in set (0.00 sec)
```


Q4.A3: RIGHT JOIN - all products with orders (incl. never ordered)

```
Q4.A3: RIGHT JOIN - all products with orders (incl. never ordered)

mysql> USE q4_ecommerce;
Database changed

mysql> SELECT p.prod_name, o.order_id
-> FROM ORDER_ITEMS oi
-> RIGHT JOIN PRODUCT p ON oi.prod_id = p.prod_id
-> LEFT JOIN ORDERS o ON oi.order_id = o.order_id
-> ORDER BY p.prod_name;
+-----+-----+
| prod_name | order_id |
+-----+-----+
| Backpack  | 5        |
| Backpack  | 8        |
| Bedsheet  | 4        |
| Bedsheet  | 8        |
| Cookware Set | 4        |
| Cookware Set | 7        |
| Headphones | 1        |
| Headphones | 3        |
| Headphones | 9        |
| Laptop    | 1        |
| Laptop    | 6        |
| Office Chair | 2        |
| Office Chair | 7        |
| Running Shoes | 5        |
| Running Shoes | 9        |
| Smartphone | 3        |
| Smartphone | 6        |
| Smartphone | 10       |
| Smartwatch | NULL     |
| Study Table | 2        |
| Study Table | 10       |
+-----+-----+
21 rows in set (0.00 sec)
```

Q4.A4: Top-selling product category by total revenue

```
Q4.A4: Top-selling product category by total revenue

mysql> USE q4_ecommerce;
Database changed

mysql> SELECT p.category, SUM(oi.quantity * p.price) AS total_revenue
-> FROM ORDER_ITEMS oi
-> JOIN PRODUCT p ON oi.prod_id = p.prod_id
-> GROUP BY p.category
-> ORDER BY total_revenue DESC;
+-----+-----+
| category | total_revenue |
+-----+-----+
| Electronics | 200000 |
| Furniture   | 31500  |
| Home        | 18000   |
| Fashion     | 11400   |
+-----+-----+
4 rows in set (0.00 sec)
```

Q4.B1: VIEW - ORDER_SUMMARY (total_amount via join + SUM)

```
Q4.B1: VIEW - ORDER_SUMMARY (total_amount via join + SUM)

mysql> USE q4_ecommerce;
Database changed

mysql> DROP VIEW IF EXISTS ORDER_SUMMARY;
Query OK, 0 rows affected (0.00 sec)

mysql> CREATE VIEW ORDER_SUMMARY AS
-> SELECT o.order_id, c.cust_name, o.order_date,
->        SUM(oi.quantity * p.price) AS total_amount
-> FROM ORDERS o
-> JOIN CUSTOMER c ON o.cust_id = c.cust_id
-> JOIN ORDER_ITEMS oi ON o.order_id = oi.order_id
-> JOIN PRODUCT p ON oi.prod_id = p.prod_id
-> GROUP BY o.order_id, c.cust_name, o.order_date;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM ORDER_SUMMARY ORDER BY order_id;
+-----+-----+-----+-----+
| order_id | cust_name | order_date | total_amount |
+-----+-----+-----+-----+
| 1 | Rahul Bansal | 2026-01-02 | 60000 |
| 2 | Rahul Bansal | 2026-01-15 | 10500 |
| 3 | Sneha Kapoor | 2026-01-05 | 27500 |
| 4 | Vikas Chauhan | 2026-01-08 | 10000 |
| 5 | Vikas Chauhan | 2026-01-20 | 6600 |
| 6 | Priyanka Rao | 2026-01-10 | 80000 |
| 7 | Priyanka Rao | 2026-01-25 | 15200 |
| 8 | Meghna Iyer | 2026-01-12 | 6600 |
| 9 | Sneha Kapoor | 2026-01-28 | 10500 |
| 10 | Meghna Iyer | 2026-01-30 | 34000 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

Q4.B2: VIEW - CUSTOMER_SPENDING (GROUP BY inside view)

```
Q4.B2: VIEW - CUSTOMER_SPENDING (GROUP BY inside view)

mysql> USE q4_ecommerce;
Database changed

mysql> DROP VIEW IF EXISTS CUSTOMER_SPENDING;
Query OK, 0 rows affected (0.00 sec)

mysql> CREATE VIEW CUSTOMER_SPENDING AS
-> SELECT c.cust_name, COALESCE(SUM(oi.quantity * p.price), 0) AS total_spent
-> FROM CUSTOMER c
-> LEFT JOIN ORDERS o ON c.cust_id = o.cust_id
-> LEFT JOIN ORDER_ITEMS oi ON o.order_id = oi.order_id
-> LEFT JOIN PRODUCT p ON oi.prod_id = p.prod_id
-> GROUP BY c.cust_name;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM CUSTOMER_SPENDING ORDER BY total_spent DESC;
+-----+-----+
| cust_name | total_spent |
+-----+-----+
| Priyanka Rao | 95200 |
| Rahul Bansal | 70500 |
| Meghna Iyer | 40600 |
| Sneha Kapoor | 38000 |
| Vikas Chauhan | 16600 |
| Farhan Sheikh | 0 |
+-----+-----+
6 rows in set (0.00 sec)
```

Q4.B3: Customers who spent more than 10000, via CUSTOMER_SPENDING

●●● Q4.B3: Customers who spent more than 10000, via CUSTOMER_SPENDING

```
mysql> USE q4_ecommerce;
```

```
Database changed
```

```
mysql> SELECT cust_name, total_spent
```

```
-> FROM CUSTOMER_SPENDING
```

```
-> WHERE total_spent > 10000
```

```
-> ORDER BY total_spent DESC;
```

```
+-----+-----+
| cust_name | total_spent |
+-----+-----+
| Priyanka Rao | 95200 |
| Rahul Bansal | 70500 |
| Meghna Iyer | 40600 |
| Sneha Kapoor | 38000 |
| Vikas Chauhan | 16600 |
+-----+-----+
```

```
5 rows in set (0.00 sec)
```